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RESEARCH ARTICLE

Towards improving Arctic mixed-phase cloud representation in the ECMWF model using MOSAiC observations

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Abstract

The presence of Arctic clouds plays a crucial role in the evolution of the surface temperature of Arctic sea ice. However, large biases in cloud representation remain in state-of-the-art weather and climate models. In this study, we use observational data from the one-year Arctic ship campaign Multi-disciplinary drifting Observatory for the Study of Arctic Climate (MOSAiC) to evaluate the Integrated Forecasting System (IFS), the global weather prediction model of the European Centre for Medium-range Weather Forecasts (ECMWF). We find seasonal biases in the cloud liquid water path, which is underestimated in winter and overestimated in summer. We show that the occurrence of supercooled liquid clouds with strong supercooling (below −20°C) is underestimated, while the occurrence of liquid-containing clouds close to freezing temperature is overestimated. Focusing on winter, we investigate the sensitivity of supercooled liquid clouds to different model uncertainties in a single-column model setup. We find the strongest sensitivity is to uncertainties in cloud microphysics and show that reducing the assumed ice particle number concentration (within the bounds of uncertainty) reduces the ice deposition rate and improves the underestimation of supercooled liquid clouds significantly. As a result, the long-wave downward radiation at the surface improves. Positive effects on the boundary-layer structure are shown in a case study. Furthermore, we document the sensitivity of clouds to ice-particle fall speed and to small fractions of open ocean in the sea ice, which are often present in the model when satellite observations show full sea-ice cover. Both sensitivities are minor compared with the sensitivity to uncertain assumptions that affect the ice deposition rate directly. Our results can guide future model development to reduce Arctic cloud biases.

KEYWORDS

Arctic, microphysics, mixed-phase clouds, MOSAiC, numerical weather prediction, sea-ice concentration, winter

1 INTRODUCTION

The Arctic is a region especially sensitive to climate change, most visible in rapid sea-ice loss, which increases the interest in predicting weather and sea-ice conditions in the Arctic ocean (Jung *et al.*, 2016). Low-level mixed-phase clouds are a key component of the Arctic climate system (e.g., Tan & Storelvmo, 2019). They occur frequently and their ability to reflect and emit radiation impacts the surface energy budget (e.g., Shupe & Intrieri, 2004). The radiative impact of clouds dominates the variability of the Arctic surface energy budget in winter (Stramler *et al.*, 2011; Tao *et al.*, 2025). However, models struggle to predict the occurrence of mixed-phase clouds (Cesana *et al.*, 2012; Duffey *et al.*, 2025; Pithan *et al.*, 2014), because their persistence depends on a delicate balance of parametrised processes (e.g., Morrison *et al.*, 2012; Morrison *et al.*, 2020) and few observations exist, especially in the high Arctic and in winter.

The Multi-disciplinary drifting Observatory for the Study of Arctic Climate (MOSAiC) campaign provides a comprehensive dataset of liquid-bearing (liquid or mixed-phase) clouds in the "New Arctic" through the full annual cycle (Shupe & Rex, 2022). The MOSAiC expedition was a year-long drift with the sea ice across the Arctic Ocean aboard icebreaker *Polarstern* (Knust, 2017) from October 2019–October 2020. Ground-based and in situ observations in the Arctic ocean are rare, especially in the cold seasons. Since the last year-long Arctic ship campaign Surface Heat Budget of the Arctic Ocean (SHEBA: Uttal *et al.*, 2002), the Arctic has warmed by about 2°C (Buontempo *et al.*, 2022)[1] and is undergoing a transition from thick multiyear ice to thin first-year ice (Kwok, 2018), marking a substantial change thus referred to as the "New Arctic". This "New Arctic" has not been probed in winter before, providing a unique opportunity to evaluate how models represent the changed Arctic climate system.

The European Centre for Medium-range Weather Forecasts (ECMWF) Integrated Forecasting System (IFS) is a state-of-the-art atmospheric model used for operational weather forecasts as well as high-resolution climate projections (Haarsma *et al.*, 2020; Rackow *et al.*, 2025). In addition, the IFS is the basis model for the widely used ECMWF Reanalysis Version 5 (ERA5) product (Hersbach *et al.*, 2020), and systematic model errors can affect the quality of the reanalysis. Despite efforts to improve the representation of high-latitude clouds (Forbes & Tompkins, 2011), problems in the representation of Arctic clouds in different model versions of the IFS remain. Sotiropoulou *et al.* (2016), Tjernström *et al.* (2021), and McCusker *et al.* (2023) report an overestimate of the occurrence of liquid-bearing clouds and an underestimate of clear-sky periods in summer. In contrast, Engström *et al.* (2014) and Solomon *et al.* (2023) report an underestimate of the occurrence of liquid-bearing clouds in winter. However, different model versions have been evaluated in different seasons, preventing a consistent evaluation across seasons, and no sensitivity tests with more recent versions of the IFS have been performed.

The underestimation of liquid water in Arctic mixed-phase clouds is a common issue in models, and various studies have found a strong sensitivity to ice microphysics, notably the assumed number concentration of ice particles (e.g., Gjelsvik *et al.*, 2025; Klein *et al.*, 2009; Morrison & Pinto, 2006; Pinto, 1998; Prenni *et al.*, 2007; Solomon *et al.*, 2009). At temperatures above −38°C, ice nucleating particles (INP) are needed to initiate freezing, and the Arctic generally has lower INP concentrations compared with lower latitudes (e.g., Bigg, 1996; Gjelsvik *et al.*, 2025; Prenni *et al.*, 2007; Schrod *et al.*, 2020; Welti *et al.*, 2020). However, the IFS model uses an ice nucleating particle parametrisation that depends solely on temperature (Meyers *et al.*, 1992), without accounting for varying aerosol conditions. Furthermore, the parametrisation used is based on measurements between −20°C and −7°C, and cannot be assumed to be valid at temperatures below that.

Several parametrisations of varying complexity have been proposed to connect ice nucleating particle concentration to aerosol concentration (e.g., DeMott *et al.*, 2010, 2015; McCluskey *et al.*, 2018; Niemand *et al.*, 2012). A lower ice particle number concentration reduces the efficiency of water-vapour deposition on ice particles, helping to maintain liquid cloud water. While Engström *et al.* (2014) found that reducing the deposition rate via a simple scaling increases the occurrence of supercooled liquid water in Arctic winter in an older version of the IFS, no attempts have been undertaken to set the assumed INP concentration in the IFS deposition rate based on Arctic aerosol conditions.

Another aspect of Arctic low-level clouds in winter is the sensitivity to sea-ice conditions. The temperature contrast between the warm ocean and the cold sea-ice surface is large and has the potential to affect cloud formation by evaporation of ocean water and change in the near-surface atmospheric stability. Saavedra Garfias *et al.* (2023) report that clouds downwind of sea-ice leads have increased total liquid water content based on MOSAiC observations. Lüpkes *et al.* (2008) report that small changes in sea-ice concentration above 90% have a strong impact on near-surface temperature in Arctic winter in clear-sky conditions. In high sea-ice concentration conditions (as present in winter), satellite retrievals have an uncertainty of up to 5% (Spreen *et al.*, 2008). The impact of small variations in sea-ice concentration within this uncertainty on clouds in central Arctic winter in the IFS has, to our knowledge, not been documented yet.

Using single-column model (SCM) versions of global models is useful for exploring model sensitivities to uncertain physical parametrisations, as they are computationally inexpensive to run and the parametrisations affect only one column. Several studies have used SCMs to examine cloud parametrisations (e.g., Engström *et al.*, 2014; Klaus *et al.*, 2012). SCM studies are particularly well-suited for evaluation against ground-based (shipborne or station) data, as both the simulation and observational data are columnar in nature. However, it is important to test the representativeness of the SCM setup (e.g., the influence of forcing) to ensure that the sensitivity results are transferable to the 3D model.

In this study, we use ground-based remote sensing and radiosonde data from the MOSAiC campaign to evaluate the occurrence of liquid-bearing clouds in the recent IFS model version through the full annual cycle. Focusing on winter conditions, which are covered least by in situ/ground-based data, we perform sensitivity tests to highlight relevant processes that could lead to observed systematic cloud errors.

We address the following research questions,

- How well can a recent version of the IFS predict the occurrence of liquid-bearing clouds in the central Arctic through the full annual cycle? Are previously reported model biases reproduced and can we identify any dependences of liquid-bearing cloud errors on temperature (Section 3.1)?
- Can we use aerosol information from MOSAiC in the depositional ice growth parametrisation (deposition rate) to reduce the underestimate of liquid-bearing clouds in winter, and what is the impact of such microphysics changes on boundary-layer profiles and broad-band radiation at the surface (Section 3.2)?
- How strong is the sensitivity of model clouds under winter conditions to other uncertainties in the model, such as the ice-particle fall speed (Section 3.3.1) or small fractions of open ocean within the sea ice (Section 3.3.2)?

We describe relevant aspects of the model parametrisation, model setup, and observational data in Section 2. The results are presented in Section 3 and discussed in Section 4 and conclusions are summarised in Section 5.

2 | METHODS

This section has two parts describing the model and the observational data used in this study. The model part (Section 2.1) describes the relevant physical parametrisations (Section 2.1.1) used in both the global model and the single-column model setup (Section 2.1.2). The data part (Section 2.2) summarises the observations from the MOSAiC campaign used for the evaluation.

2.1 | The model

We evaluate forecasts of the ECMWF IFS run using IFS model version CY48R1 (operational at ECMWF from July 2023–November 2024), initialised from the operational analysis and run at a grid resolution of around 18 km (TCo639) with 137 vertical levels. Vertical levels are closely spaced near the surface, with a layer depth of around 20 m close to the surface and around 120 m at the typical height of low-level cloud (about 1 km). The IFS is coupled to the Nucleus for European Modelling of the Ocean (NEMO) ocean model V3.4 and the version two of Louvain-la-Neuve sea ice model (LIM2) sea-ice model (Mogensen *et al.*, 2012) at a 0.25° resolution and initialised from the ECMWF OCEAN5 analysis (Zuo *et al.*, 2019).

We also evaluate ERA5 (Hersbach *et al.*, 2020), which is a reanalysis dataset based on 4D-Var data assimilation and the IFS model version CY41R2. It is produced at a resolution of around 31 km (TL639) with the same 137 vertical levels and provides hourly data from short forecasts within the assimilation framework.

2.1.1 | Model parametrisations

The full physics package of the IFS model is described in the documentation (ECMWF, 2023). It includes a convection parametrisation (Bechtold *et al.*, 2008; Bechtold *et al.*, 2014). The radiation scheme is described in Hogan and Bozzo (2018). Subgrid turbulent mixing is parametrised using an eddy-diffusivity mass-flux (EDMF) framework described in Siebesma *et al.* (2007) and Köhler *et al.* (2011). Developments to improve the turbulent mixing in stable conditions are described in Sandu *et al.* (2014). The subgrid cloud and precipitation parametrisation is based on Tiedtke (1993), featuring a prognostic cloud fraction variable and single-moment microphysics. Substantial upgrades have been made to the cloud scheme over the years, including prognostic variables for the bulk mass of four separate hydrometeor types (cloud water, cloud ice, rain, snow) and an improved representation of microphysical processes (Forbes & Ahlgrimm, 2014; Forbes *et al.*, 2011; Forbes & Tompkins, 2011).

Sources and sinks of cloud liquid water in the mixed-phase regime

In the mixed-phase temperature range (−38 °C to 0 °C), all condensate that is produced through the air reaching saturation (large-scale lifting, radiative cooling, vertical subgrid mixing, and transport by turbulence) goes into the liquid condensate category. In the presence of water, the air will be at water saturation and hence supersaturated with respect to ice. The nucleation and depositional growth of ice in this supersaturated environment drive the phase partitioning between liquid and ice and are described in more detail in the next section. Other sinks of cloud liquid water are autoconversion and accretion to rain, and riming on snow.

Arctic mixed-phase clouds typically have a thin layer of liquid water at the top, with ice particles nucleating, growing, and precipitating out (Morrison *et al.*, 2012). The liquid water content is too low for effective rain formation and, as the ice particles fall out of the liquid layer, there is not sufficient time for riming to occur. Therefore, the depositional growth process is the main process reducing the cloud liquid water in Arctic mixed-phase boundary-layer clouds.

Depositional ice growth

The deposition process is the main sink of cloud liquid water in mixed-phase clouds in the IFS. It implements the ice growth at the expense of liquid water via the water-vapour phase because of the lower saturation vapour pressure over ice than over water at subzero temperatures (WBF process: Wegener, 1911; Bergeron, 1935; Findeisen, 1938). The liquid water content is reduced and the ice water content is increased directly, without explicitly simulating the transition via the water-vapour phase, because the evaporation/condensation of liquid water is expected to be fast compared with the model time step (7 min 30 s). We first describe the current operational configuration and then describe how we added the dependence on aerosol concentrations.

The ice deposition parametrisation describes the diffusion of water vapour onto ice particles and depends on various assumptions: the difference in water-vapour pressure and saturation vapour pressure with respect to ice drives the process; as long as liquid water is present in the cloud, the air is assumed to be at water saturation; the process rate depends on the ice-particle capacitance C and number concentration N_i; all particles are assumed to be the same size, that is, a monodisperse size spectrum. The implementation in the IFS follows (Rotstayn *et al.*, 2000) and describes the change of the specific ice water content dq_i with time step dt by

$$\frac{dq_\mathrm{i}}{dt} = \frac{N_\mathrm{i}}{\rho}\frac{4\pi C}{A'' + B''}\frac{e_\mathrm{sl} - e_\mathrm{si}}{e_\mathrm{si}}, \tag{1}$$

where A'' and B'' represent heat conduction and vapour diffusion, respectively, and e_sl and e_si the saturation water-vapour pressure with respect to liquid and to ice, respectively.

The capacitance C depends on the ice-particle shape and is assumed to be $C = D_\mathrm{i}/2$, where D_i is the diameter of the ice particle, which is an assumption valid for spherical particles. Assuming that particles are monodispersed and follow a mass diameter relation ($M_\mathrm{i} = a_\mathrm{i} D_\mathrm{i}^{b_\mathrm{i}}$), we can formulate the ice-particle diameter D_i in terms of the ice-mass mixing ratio q_i and the assumed number concentration N_i:

$$D_\mathrm{mono} = \left(\frac{M_\mathrm{i}}{a_\mathrm{i}}\right)^{1/b_\mathrm{i}} = \left(\frac{q_\mathrm{i}\rho}{a_\mathrm{i}N_\mathrm{i}}\right)^{1/b_\mathrm{i}}. \tag{2}$$

For spherical particles, the mass–diameter relation is given as

$$M_\mathrm{i} = \frac{\rho_\mathrm{i}\pi}{6}D_\mathrm{i}^3,$$

where ρ_i is the particle density, which leads to the deposition-rate formulation used in the operational IFS:

$$\frac{dq_\mathrm{i}}{dt} = \left(\frac{N_\mathrm{i}}{\rho}\right)^{2/3}\left(\frac{48\pi^2}{\rho_\mathrm{i}}\right)^{1/3}\frac{q_\mathrm{i}^{1/3}}{(A'' + B'')}\frac{e_\mathrm{sl} - e_\mathrm{si}}{e_\mathrm{si}}. \tag{3}$$

The number concentration of ice particles (m^{-3}) is diagnosed using the INP concentration parametrisation by Meyers *et al.* (1992) (hereafter Meyers), which is based on the relative difference of saturation vapour pressure with respect to liquid and ice, which depends solely on temperature:

$$N_\mathrm{i} = 1000\exp\left(12.96\frac{e_\mathrm{sl} - e_\mathrm{si}}{e_\mathrm{si}} - 0.639\right). \tag{4}$$

The deposition rate (Equation 3) depends on ice mass, so it needs some initial ice mass to be active. The heterogeneous nucleation of ice particles is assumed to be active below a temperature of −5°C whenever supercooled liquid water is present, but is represented simply as an initial ice mass given by the diagnostic INP number concentration and an assumed nucleated mass of each ice particle of 1×10^{-12} kg/kg.

The deposition rate has a multiplication factor of 0.65, representing the reduction in deposition rate due to supercooled liquid water and ice not fully overlapping spatially. Spatial separation of liquid and ice has been observed from aircraft but is not well quantified (Korolev & Milbrandt, 2022), and this parameter was previously chosen to give the best average global impacts in the operational model.

In addition, because of limited vertical resolution, large-scale models cannot resolve the vertical separation of cloud liquid water and ice sufficiently, which leads to an overestimation of the deposition rate at the cloud top (Barrett *et al.*, 2017). To account for this overestimation, the deposition rate is reduced at the cloud top. The simple implementation of the deposition-rate reduction at the cloud top is described in Forbes and Tompkins (2011), although the impact is relatively small.

Adding an aerosol dependence to the deposition parametrisation

The deposition rate is sensitive to the assumed ice particle number concentration, which is diagnosed using an INP concentration parametrisation. The operational setup uses a temperature-dependent parametrisation by Meyers *et al.* (1992). This dependence on temperature alone is a simplification, as observations have shown very different INP concentrations in different environments (e.g., Welti *et al.*, 2020), and several parametrisations that link INP concentration to aerosol concentrations have been developed (e.g., DeMott *et al.*, 2010, 2015, 2016; McCluskey *et al.*, 2018; Niemand *et al.*, 2012; Phillips *et al.*, 2013). DeMott *et al.* (2010) (hereafter DM10) suggest a comparatively simple parametrisation based on multiple field campaigns that predicts the INP concentration based on temperature and the concentration of coarse aerosol ($> 0.5\ \mu$m) and shows a better skill in predicting INP concentrations compared with solely temperature-dependent parametrisations.

In this study, we test the use of the DM10 parametrisation to diagnose INP concentrations based on temperature and aerosol concentration:

$$N_{\mathrm{i}}(T_k, n_{\mathrm{aer},0.5}) = a(273.16 - T_k)^b (n_{\mathrm{aer},0.5})^{(c(273.16-T_k)+d)}, \quad (5)$$

where $a = 0.0594$, $b = 3.33$, $c = 0.0264$, $d = 0.0033$, T_k is the cloud temperature in Kelvin, $n_{\mathrm{aer},0.5}$ is the number concentration (cm^{-3}) of aerosol particles with diameters larger than 0.5 μm, and N_{i} is the ice nuclei number concentration (m^{-3}) at T_k. The DM10 parametrisation is valid (based on observations) in the temperature range from −35°C to −9°C. The sensitivity of Arctic winter clouds to the assumed INP concentration based on DM10 and observed levels of aerosol concentration is described in Section 3.2.

2.1.2 | Single-column setup for sensitivity tests

In the 3D model, parametrisations act independently in each vertical column, so the single-column model is a suitable tool for testing the local impact of changes to parametrisations in a computationally efficient way. We use the IFS single-column model with model version 48R1. The single-column model needs prescribed initial conditions and forcing to represent horizontal advection and large-scale vertical motion. The SCM forcing is derived from two-day coupled short-range forecasts with the IFS (cycle 48R1) at a resolution of around 25 km (TCo399). Every 3 hours, atmospheric profiles and surface fields are extracted at the model grid column closest to the *Polarstern* position at 0000 UTC, updating the location daily. Advective tendencies of temperature and humidity profiles are derived from local gradients and winds.

The SCM is run daily, with advective tendencies of humidity and temperature applied, strong wind nudging (1-hour timescale), and weak nudging of temperature and humidity (48-hour timescale). The forcing data are interpolated linearly between the three-hourly time steps. This setup with daily initialisation and a weak temperature and humidity nudging allows the physical parametrisations to modify the profile while preventing a drift too far from the atmospheric state of the forcing. We evaluate day 2 (24–48 h forecast) of the SCM simulations, comparable with the 3D model evaluation. Sea-ice fraction is also initialised and updated every 3 hours with linear interpolation in time, using the data from the IFS, which are coupled with the NEMO ocean model and LIM3 sea-ice model.

2.2 | The observational data

The MOSAiC campaign provided surface-based observations in the central Arctic for a full year. The following data are used for evaluation. Some of the data are available as merged observatory data files (MODFs), which facilitates the process of model evaluation (Uttal *et al.*, 2024).

The main diagnostic quantity used here for the evaluation of liquid-containing clouds is the vertically integrated cloud liquid water (liquid water path, LWP) retrieval from the Humidity And Temperature PROfilers (HATPRO) by Walbröl *et al.* (2022). The LWP retrieval is not affected by the presence of ice clouds and has an uncertainty of 14–23 g · m^{-2}, lowest for low LWPs and highest for high LWPs.

For evaluation of cloud profiles we use the multi-instrument retrieval of cloud liquid and ice water content by Shupe *et al.* (2015). Reported uncertainty for ice water content is a multiplicative factor of 2, and the uncertainty for liquid water content is determined by the LWP uncertainty and assumptions on liquid cloud base and top height (Shupe *et al.*, 2015). The evaluation of temperature and humidity profiles is based on the blended dataset from radiosonde and meteorological tower data (Dahlke *et al.*, 2023).

The cloud-base temperature is diagnosed based on the interpolated radiosonde product (Jensen *et al.*, 2020) and the *Polarstern* ceilometer (Schmithüsen *et al.*, 2021). Broad-band downwelling long-wave radiation at the surface (LW down) and surface latent heat flux are evaluated using MetCity observations taken in the vicinity of the ship (Cox *et al.*, 2023). The average uncertainty of LW down radiation fluxes is 2.6 $W \cdot m^{-2}$. Latent heat fluxes are measured using the eddy covariance method; a robust assessment of the uncertainty is still subject to scientific challenge and comparison with the latent heat flux data should be taken with care. It also needs to be taken into account that the latent heat flux does not necessarily incorporate the effect of leads, because it is based on measurements by the tower, which is placed where there is full sea-ice cover.

For comparison of the sea-ice fraction, we use the retrieved dataset described in Krumpen *et al.* (2021), based on satellite microwave radiometers Advanced Microwave Scanning Radiometer-EOS (AMSR-E) and Advanced Microwave Scanning Radiometer 2 (AMSR2). The reported uncertainty is below 5% for high ice concentration regimes in winter and up to 25% in summer (Spreen *et al.*, 2008).

The model parameter choice for INP is informed by coarse aerosol concentrations measured aboard *Polarstern* with the Wideband Integrated Bioaerosol Sensor (WIBS: Beck *et al.*, 2023), which records hourly aerosol particle number concentration between 0.5 and 20 μm in optical diameter, as described by Heutte *et al.* (2023). Observational data used for evaluation is publicly available and summarised in Table 2.

3 | RESULTS

3.1 | Representation of (supercooled) liquid cloud water

This section examines how well the IFS captures the annual cycle of the occurrence and magnitude of cloud liquid water. We evaluate day 2 (step 24–48 h) of forecasts initialised daily from the operational IFS analyses at 0000 UTC at the grid point nearest to *Polarstern*'s position at initialisation time, comparable with the evaluation setup from Solomon *et al.* (2023). Due to a data processing error, model data from day 28.07.2020 are missing and hence excluded from the analysis. Temperature, humidity, and wind data from radiosondes launched during the MOSAiC campaign were assimilated into the operational analyses. The initial profile is therefore close to the observed state and we thereby assess fast-growing errors from the parametrisations. Figure 1 shows the LWP from the model and the observations for the full MOSAiC drift period.

3.1.1 | Annual cycle

Over the full MOSAiC time period, the observations show a distinct annual cycle in the occurrence and magnitude of cloud liquid water. From late November to April, the atmosphere is predominantly free of liquid-containing clouds (LWP > 20 $g \cdot m^{-2}$), with only occasional liquid-containing cloudy periods ranging in duration from half a day to multiple days. This persistence of liquid-containing clouds in the Arctic is a known phenomenon (e.g., Morrison *et al.*, 2012). In contrast, from May to late September, liquid-containing clouds are almost always present, apart from a few cloud-free days.

The IFS reproduces the general seasonal pattern, with many liquid water-containing clouds in summer and few liquid water-containing clouds in winter. However, the model exaggerates this seasonal contrast. The seasonal distributions of LWP in the model and observations are shown in Figure 2. In winter (December–January–February, DJF), almost no clouds containing liquid water are predicted; consequently, their occurrence is underestimated significantly in the model. In summer (June–July–August, JJA), the model is too cloudy and misses the liquid cloud-free days observed. In autumn (September–October–November, SON) and spring (March–April–May, MAM) the LWP distributions are represented reasonably well.

Through the annual cycle, the atmospheric air temperature changes and with it the degree of supercooling of occurring cloud liquid water, which is critical for the efficiency of microphysical processes that convert liquid water to ice. In the model, the primary sink process for supercooled cloud liquid water—the deposition rate—is strongly dependent on temperature (see Section 2.1.1 for a more detailed description).

3.1.2 | Temperature dependence

To evaluate the temperature dependence, we examine the cloud-base temperature from both observations and the model as an indicator of the temperature regime of the clouds. Because of the small depth of the liquid layer (Morrison *et al.*, 2012), the cloud-base temperature provides a reasonable estimate of the in-cloud temperature of the (lowest) cloud layer. By analysing the occurrence of clouds within specific temperature ranges, we aim to identify potential weaknesses in the model's representation of the temperature dependence of cloud microphysical processes.

To determine the liquid cloud-base temperature from the observations, we first filter for time steps where the measured LWP is above the averaged observation uncertainty of 20 $g \cdot m^{-2}$. The cloud-base height is determined by the ceilometer, which is sensitive to the liquid cloud base and not precipitating ice (e.g., Maturilli & Ebell, 2018). Outliers above 10 km are removed (five

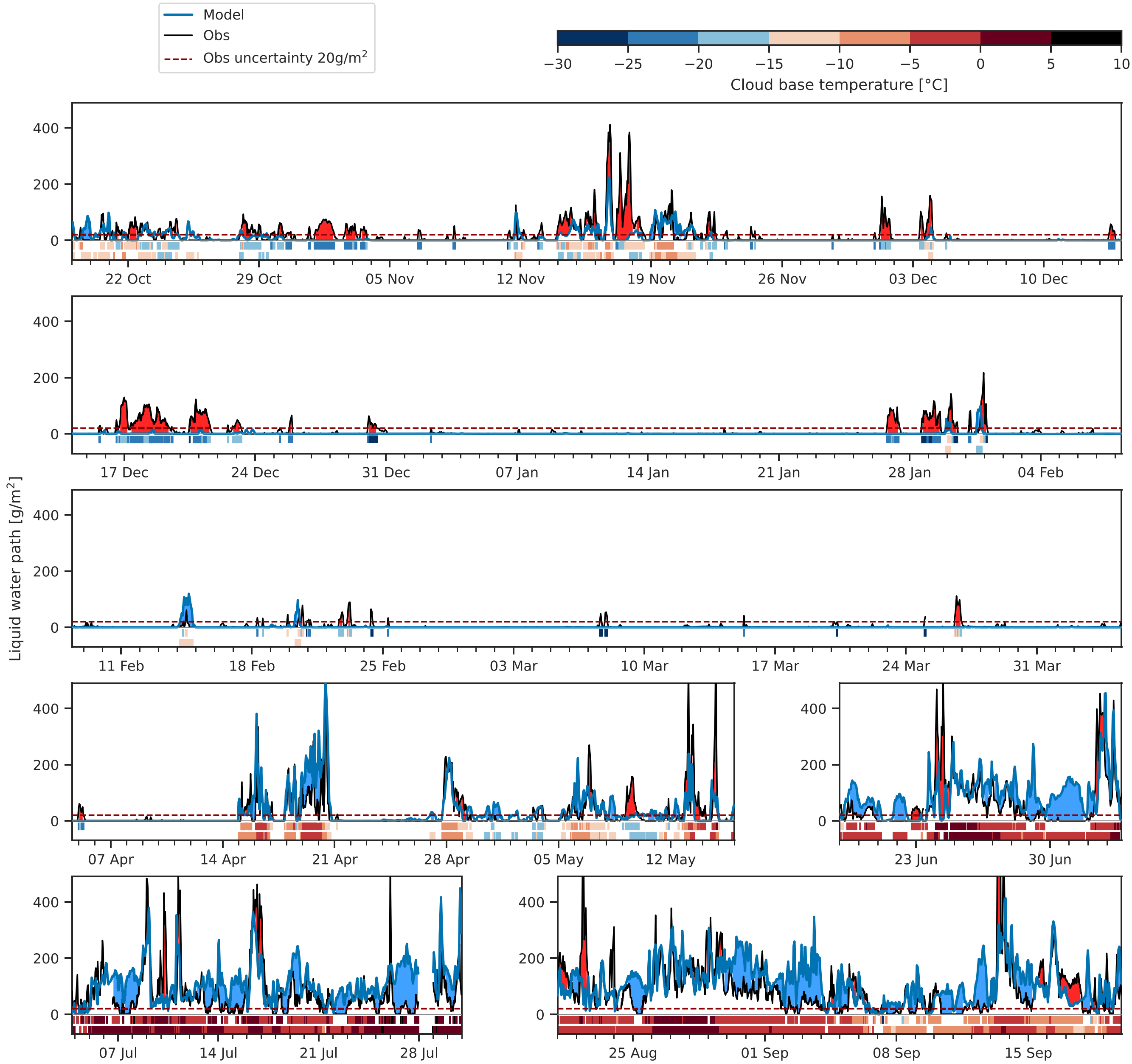

FIGURE 1 Time series of hourly averages of vertically integrated liquid water (liquid water path, LWP) in model and observations from MOSAiC. Model LWP is shown with solid blue lines, observed LWP is shown with solid black lines. The difference between observed and modelled LWP is highlighted by colour shading: model underestimates LWP (red) and model overestimates LWP (blue). Colour bars indicate cloud-base temperature from observations (upper) and cloud-base temperature from model (lower).

data points in July). To get the cloud-base temperature, we extract the temperature from the time-interpolated radiosonde profiles (Jensen *et al.*, 2020) at cloud-base height. To determine the cloud-base temperature in the model, we similarly filter for cloudy time steps (LWP > $20\ \mathrm{g \cdot m^{-2}}$) and extract the temperature at the lowest model level exceeding a liquid water mixing ratio of $0.01\ \mathrm{g \cdot kg^{-1}}$. To match observations and model times, we only use times where all required observations are available and average to hourly values.

To relate the error in LWP to the cloud temperature, the cloud-base temperature for model and observations is shown below the LWP time series in Figure 1. As expected, the cloud-base temperature shows a seasonal cycle. In the winter, liquid-containing clouds that are missed by the model have a cold cloud-base temperature, predominantly below −15 °C, while summer clouds, which are overestimated in LWP by the model, have warmer cloud-base temperatures above −5 °C.

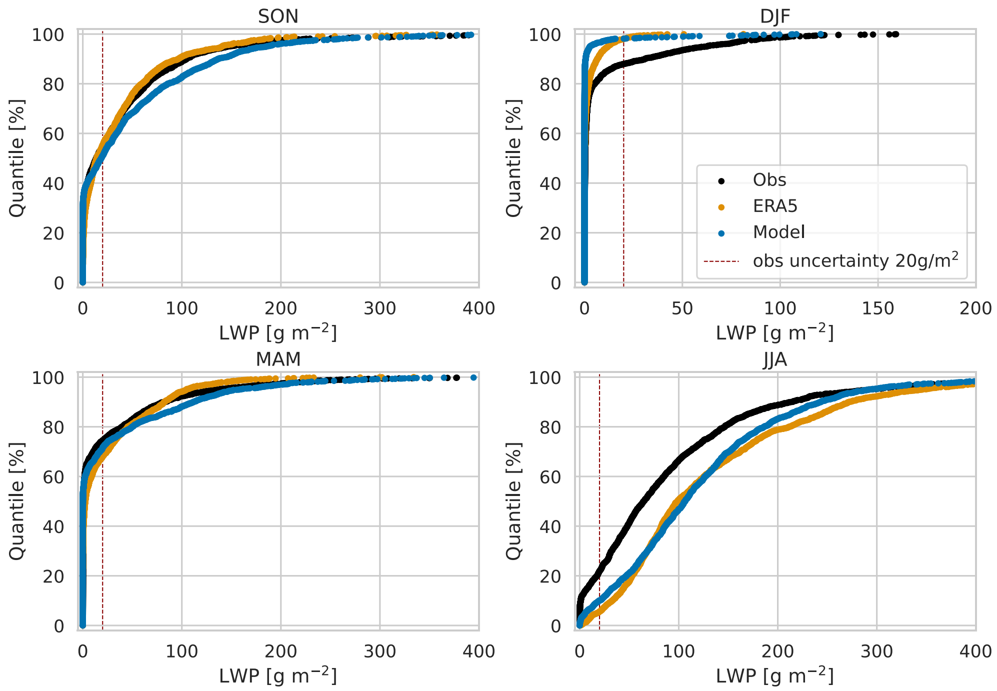

FIGURE 2 Seasonal distributions for hourly averaged LWP from observations, reanalysis ERA5, and short model forecasts. The uncertainty in observed LWP of 20 g · m^{-2} is indicated by a dashed vertical line. Note the differing *x*-axis scale for the seasons. [Colour figure can be viewed at wileyonlinelibrary.com]

Figure 3a shows how frequently liquid-containing clouds occur within different cloud-base temperature regimes in observations and in the model. The model clearly underestimates the occurrence of liquid-containing clouds with cold cloud-base temperatures. The absence of liquid-containing clouds with cloud-base temperatures below −20°C in the model could therefore be driven by an overestimated efficiency of the deposition rate at low temperatures. We test whether the model is able to produce more liquid water with a reduced deposition rate in a sensitivity experiment in Section 3.2.

The model overestimates the occurrence of liquid-containing clouds with warmer cloud-base temperature, especially for temperatures above 0°C. At temperatures above 0°C, we have no freezing, so the (uncertain) mixed-phase microphysical processes are not active at cloud base. Depending on the depth of the cloud layer, the cloud-top temperatures could be below 0°C. Whether this overestimation of warmer clouds is due to missing warm or mixed-phase sink processes or an overestimation of source processes is unclear at this point and would need further work, which is beyond the scope of this study.

In summary, the full year evaluation of LWP in the IFS shows opposite biases in the warm and cold seasons. The summer overestimate is in line with the in-depth analysis by Tjernström *et al.* (2021) using data from the summer Arctic Ocean 2018 expedition. They also report biases in the vertical temperature structure and mention a possible relation to the misrepresentation of clouds. In an idealised winter Lagrangian SCM intercomparison and a recent April case study from MOSAiC (Pithan *et al.*, 2016, 2023), the IFS produces liquid-containing clouds but with a relatively low LWP. The cloud temperature in these studies is around −17°C or warmer, which is above the temperature where we find that the model fails to represent any liquid clouds. The multi-model evaluation of the Arctic winter state by Solomon *et al.* (2023) using the comprehensive MOSAiC data showed that the underestimate of the cloudy winter state is a problem common to several models.

3.1.3 | ERA5 reanalysis

The ERA5 reanalysis, which is based on an older version of the IFS, generally shows a very similar seasonal LWP distribution (Figure 2) and has the same misrepresentation as the short model forecasts. A notable difference appears in winter (DJF), where ERA5 produces more liquid-containing clouds with low LWP (<20 g · m^{-2}) than the forecast. However, the LWP is underestimated compared with the observations.

We also tested whether the temperature-dependent misrepresentation of liquid-containing clouds occurs in the ERA5 reanalysis (Figure 3b). ERA5 also shows a clear underestimation of the occurrence of supercooled liquid water with strong supercooling (<−20 °C) and overestimates the occurrence of liquid-containing clouds around the freezing temperature, comparable with the results from the IFS forecasts. Having a temperature-dependent bias in clouds could obscure climate trend analysis, which may be performed based on this widely used reanalysis

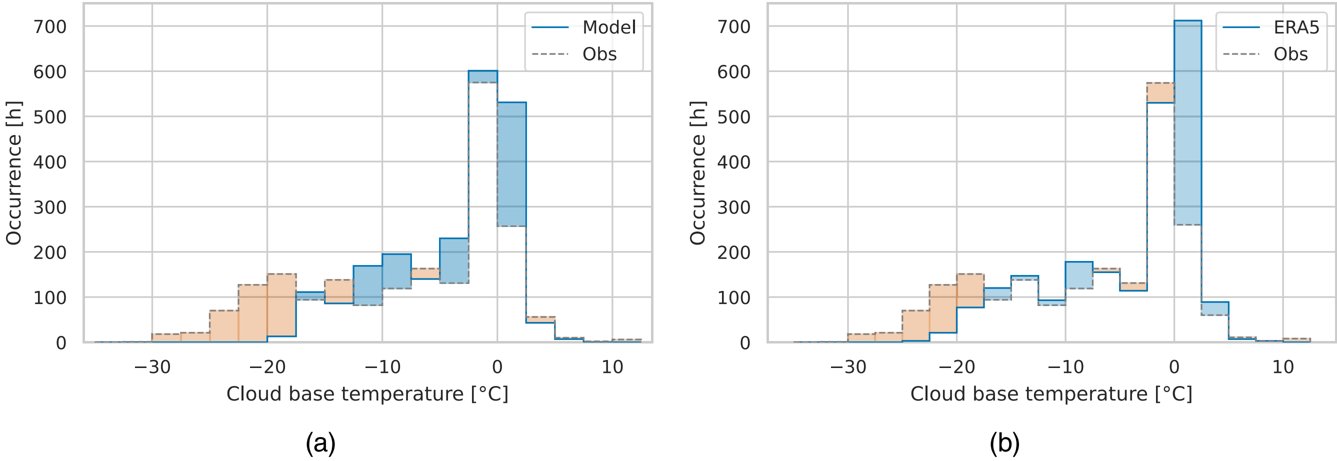

FIGURE 3 Occurrence histograms of liquid-containing clouds (hourly averaged LWP > 20 $g \cdot m^{-2}$) against cloud-base temperature for (a) model forecasts and (b) ERA5 reanalysis for the whole MOSAiC drift period (October 19, 2019–September 20, 2020). Model values are shown with solid lines, observed values with dashed lines. The difference in occurrence is highlighted by colour shading: less occurrence in the model in red, more occurrence in the model in blue. The occurrence units are hours. [Colour figure can be viewed at wileyonlinelibrary.com]

dataset. Enhancing the model's cloud representation could reduce errors in the reanalysis dataset.

It is important to note that, in winter, ERA5 has more liquid-containing clouds (LWP > 0 $g \cdot m^{-2}$) than the IFS forecast (see Figure 2). This improved representation likely results from using the analysis state rather than a forecast with a 24–48 hour lead time. Those clouds have a low LWP (< 20 $g \cdot m^{-2}$) and are thus hidden in the cloud-base temperature analysis because of the applied LWP threshold. Those clouds with low LWP in ERA5 are not sufficient to overcome the underestimate of high LW down radiation (not shown). ERA-Interim was also found to underestimate central Arctic winter LWP when compared with SHEBA (Surface Heat Budget of the Arctic Ocean) observations (Engström *et al.*, 2014), suggesting this issue persists over many model versions.

3.1.4 | Cloud-base temperature and cloud-base height

The temperature at cloud base is linked to its height: a higher cloud will have a colder cloud-base temperature, following the adiabatic lapse rate. Can the misrepresentation of cloud-base temperature be linked to a misrepresentation in cloud-base height? We investigate the relation of cloud-base temperature and height season by season (Figure 4).

In winter (DJF), observed clouds have cloud-base temperatures from −30°C to −10°C with a maximum between −22.5°C and −20°C. The model represents the occurrence frequency of warmer liquid-containing clouds (above −17.5°C) approximately, but strongly underestimates colder clouds, not representing any liquid-containing cloud below −20°C. The warmest temperature bin seems to be slightly over-represented in the model but, due to the overall low number of occurrences of liquid-containing winter clouds, it is unclear whether this difference is significant. Most clouds are observed at heights below 600 m, but higher clouds up to 3000 m cloud-base height do occur with lower frequency. The model does not represent any liquid-containing clouds close to the surface (below 100 m) and above 1200 m. Most of this misrepresentation of cloud-base heights can be explained by a temperature dependence: observed clouds at those altitudes are mostly in a very cold regime (below −20 °C) due to the cold sea-ice surface at low altitude and the afore-mentioned lapse rate for high altitude. However, in the temperature regime between −20 °C and − 10 °C the model represents only clouds below 1200 m, while clouds from the surface up to 2000 m are observed.

In summer (JJA), liquid-containing clouds occur with cloud-base temperatures predominantly between −5°C and 5°C in both observations and the model. The model overestimates the occurrence of clouds and the overestimation of clouds above 0°C is more pronounced than that below 0°C. Cloud-base height of liquid-containing clouds in both observations and the model is most frequent close to the surface (below 100 m), with frequency reducing towards higher altitudes. The model overestimates the occurrence of clouds at all altitudes, but the overestimate is most pronounced between 100 and 200 m.

In autumn (SON) and spring (MAM), the signal is more mixed. The model underestimates the occurrence

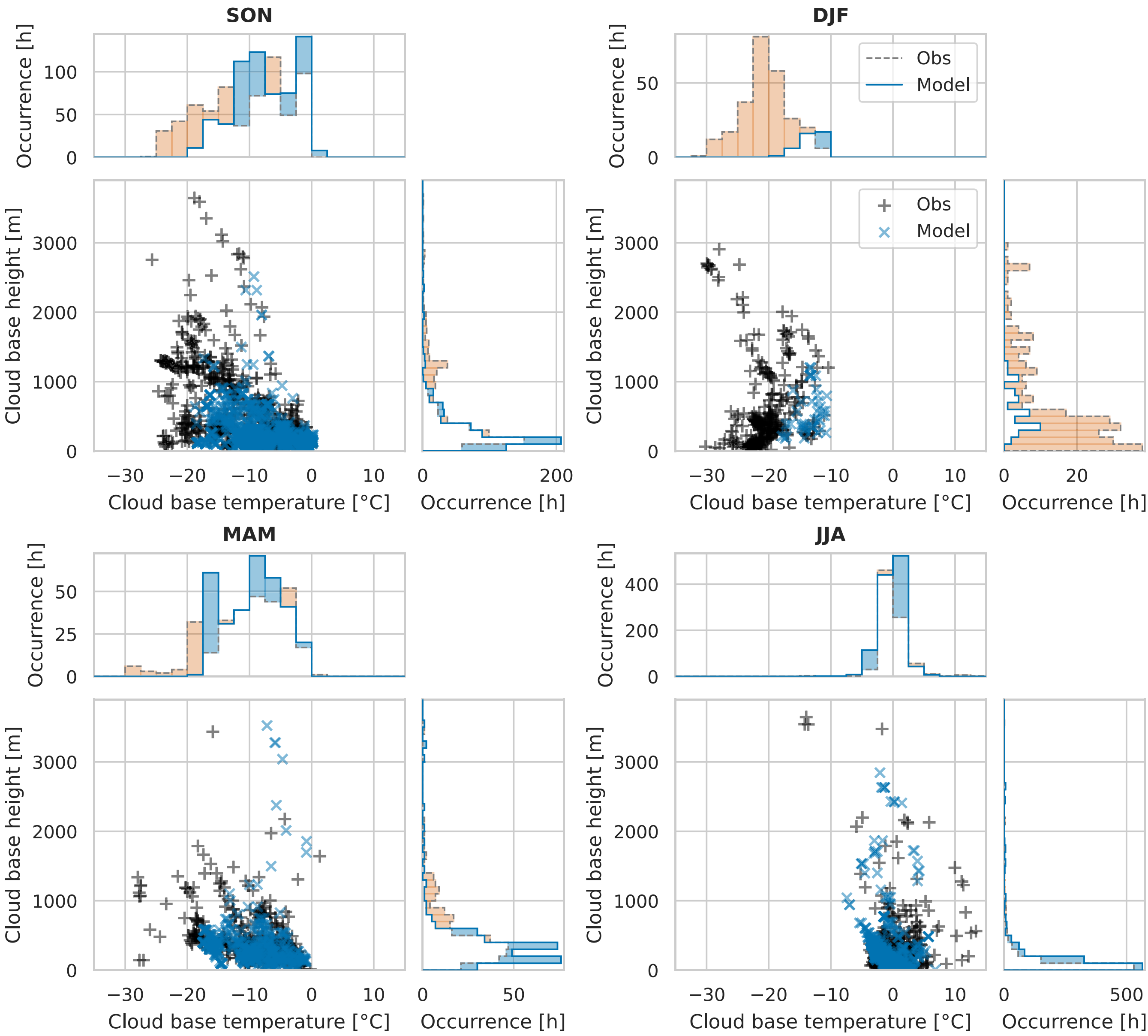

FIGURE 4 Joint distributions of the occurrence time of liquid-containing clouds (LWP>20 $g \cdot m^{-2}$) with cloud-base temperature (*x*-axis) and cloud-base height (*y*-axis). Different panels show different seasons. Model values are shown with "x" symbols (solid line) and observations with "+" symbols (dashed line) in the scatter plots (histograms). Histograms show the marginal distributions. The difference in occurrence between model and observations is highlighted by colour shading, blue (red) for an overestimated (underestimated) occurrence in the model. Note the different axis scales. [Colour figure can be viewed at wileyonlinelibrary.com]

of liquid-containing clouds with low cloud-base temperatures (below −17.5°C) and overestimates the occurrence of clouds with higher cloud-base temperatures (above −12.5°C). Furthermore, the model underestimates the occurrence of clouds with higher cloud-base height (above 1000 and 600 m in autumn and spring, respectively) and overestimates the occurrence of clouds with low cloud-base height (below 200 and 600 m in autumn and spring, respectively).

The model shows a clear temperature dependence in its cloud representation. It fails to produce any clouds at temperatures below −20°C, even though observations show that clouds do exist in these very cold conditions. This suggests a systematic deficiency in the model's ability to represent clouds in very cold environments.

The model does occasionally simulate clouds at all observed altitudes, but seems to underestimate clouds above ~ 1000 m if their temperature is below ~ −12.5°C, which is a higher temperature threshold than for lower clouds. One possible explanation for the different temperature dependence with height could be a vertical resolution dependence of the balance of sources and sinks of liquid water (Barrett *et al.*, 2017). With a coarser resolution, it takes longer to separate liquid and ice in the vertical, making it more difficult to maintain liquid water. The vertical resolution in the model increases approximately linearly with height from ∼ 20 m near the surface to ∼ 260 m above 4 km. A reduction of the vertical resolution dependence following the approach of Barrett *et al.* (2017) is planned for future model updates.

The analysis reveals a temperature dependence of the cloud error, which can help to inform the temperature dependence of parametrised processes such as the deposition rate. However, the analysis does not include occurrences of thin clouds (LWP $< 20\ \mathrm{g \cdot m^{-2}}$), as these are challenging to constrain in observations (Turner *et al.*, 2007). However, evaluation of broad-band downwelling long-wave (LW down) radiation can give insights on the radiative effect of thin clouds and will be applied in the sensitivity experiments (Section 3.2.1).

The rest of this study focuses on the representation of supercooled cloud liquid water in winter and specifically the sensitivity of modelled cloud liquid water to ice deposition rate, ice-particle fall speed, and the fractional cover of sea ice.

3.2 | Sensitivity of LWP to an aerosol-aware deposition rate in winter Arctic clouds

3.2.1 | One-month single-column model experiments

The evaluation of cloud liquid water with MOSAiC data showed a clear underestimation in the IFS in winter. In this section, we test the sensitivity of the modelled cloud liquid water to assumptions in the microphysical parametrisation to explore possibilities for model development towards a better representation of Arctic winter mixed-phase clouds.

We run the sensitivity experiments for December 2019 in a SCM version of the IFS. We choose December because frequent (supercooled) liquid-containing clouds occurred and we choose the SCM because it allows us to run experiments at low computational cost. As a check of the representativeness of the SCM experiments for the full 3D model, we run the control and one sensitivity experiment in both setups. The LWP distributions in the IFS and SCM, in control and sensitivity experiments respectively, are not identical but are sufficiently similar to justify using the SCM as an exploratory tool for model sensitivities (Figure 5d).

A limited availability of aerosol particles that act as INPs can help to maintain Arctic mixed-phase clouds and avoid glaciation (e.g., Morrison *et al.*, 2012). During MOSAiC December, coarse aerosol concentrations were observed in the range 0.1–10 $\mathrm{cm^{-3}}$ with a mean of 2.45 $\mathrm{cm^{-3}}$ (Figure 5a), measured with a WIBS (Beck *et al.*, 2023). Figure 5b shows INP concentrations predicted by the DM10 parametrisation (DeMott *et al.*, 2010) for different coarse aerosol concentrations *n_aer* (per $\mathrm{cm^3}$). Especially at low temperatures, a low aerosol concentration reduces the INP concentration significantly. As the DM10 parametrisation is based on measurements taken in the temperature regime from −9°C to −35°C, the INP concentration above −9°C is not well constrained. However, since model tropospheric air temperatures during MOSAiC December do not exceed −10°C, our results are not affected by this uncertainty. Some common temperature-dependent INP parametrisations are shown for comparison (Cooper, 1986; Meyers *et al.*, 1992; Prenni *et al.*, 2007), spanning a wide range and highlighting the uncertainty in INP parametrisations capturing different conditions.

In addition to the standard IFS (control) deposition rate using the Meyers *et al.*, (1992) INP, we implemented an aerosol-aware deposition rate (*DM10 n_aer*) based on the INP parametrisation by DeMott *et al.* (2010) and aerosol concentrations described in detail in Section 2.1.1. The INP concentration for different observed aerosol concentrations spans several orders of magnitude, which translates to large differences in the deposition rate, shown in Figure 5c. For example, at −20°C, increasing the coarse aerosol concentration by a factor of 10 results in a 3.4-fold increase in INP concentration and a 2.3-fold increase in the deposition rate (assuming an ice-mass mixing ratio of 0.1 $\mathrm{g \cdot kg^{-1}}$).

LWP distributions and liquid-containing cloud frequency (LCF)

Here we investigate the sensitivity of the modelled LWP distribution to the assumed level of aerosol load in the deposition rate. Reducing the assumed aerosol concentration to observed levels increases LWP substantially and reduces zero-LWP occurrences (Figure 5d). As a measure of liquid-containing cloud representation, we assess the relative frequency of occurrence of LWP above the observation uncertainty (20 $\mathrm{g \cdot m^2}$). This liquid-containing cloud frequency (LCF) is ∼ 22% in the observations. The LCF is clearly underestimated in the control simulation (∼ 3%). The simulation *DM10 n_aer=2*, close to the observed average coarse aerosol concentration, represents the LCF best (∼ 29%), but overestimates the occurrence of low non-zero LWP values. The simulation with a higher aerosol concentration (5 $\mathrm{cm^{-3}}$) seems to represent the number of zero-LWP cases best, but underestimates the occurrence of LWP values above ∼ 12 $\mathrm{g \cdot m^{-2}}$. The distinction between

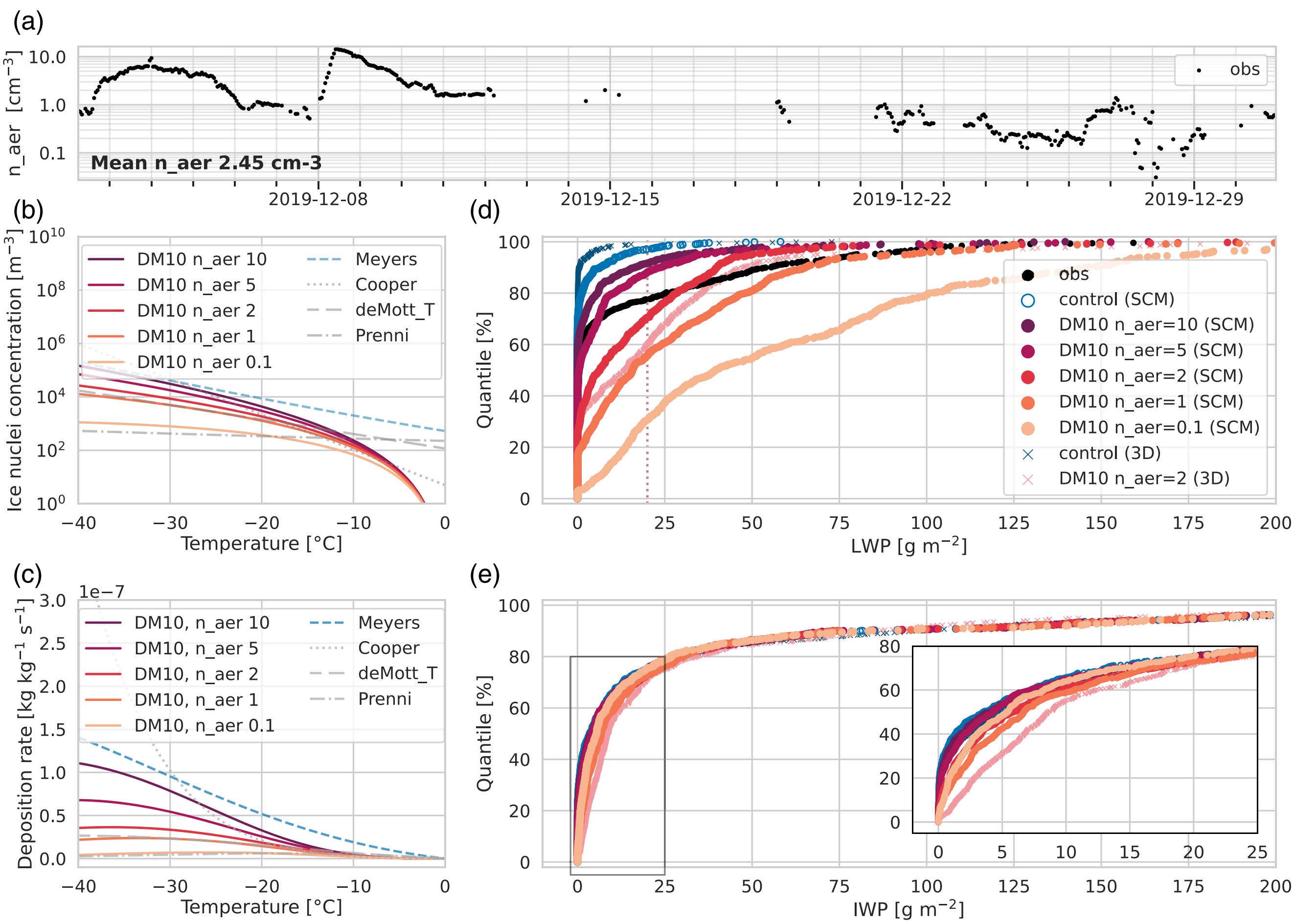

FIGURE 5 Sensitivity of the modelled LWP distribution to INP assumptions via the deposition rate. (a) Observed coarse aerosol concentrations from the Wideband Integrated Bioaerosol Sensor (WIBS); (b) INP parametrisation based on temperature and coarse aerosol concentration in cm^{-3} following DM10; (c) deposition rate for an ice-mass mixing ratio of 0.1 g · kg^{-1} and air pressure of 950 hPa using the INP concentrations from (b); (d) LWP distribution in December, comparing the observations and model simulations when using the INP concentrations following DM10. The uncertainty in observed LWP of 20 g · m^{-2} is indicated by a dotted vertical line. The value given after *n_aer* refers to the concentration of coarse aerosol particles per cm^3 used in the DM10 parametrisation. (e) IWP distribution in December from the same model simulations shown in (d). [Colour figure can be viewed at wileyonlinelibrary.com]

non-zero and zero LWP values in the observations needs to be made with care, since we compare LWP values within the uncertainty of LWP observations (20 g · m^2). When aerosol concentrations are reduced further (*DM10 n_aer=1*), the higher quantiles of the LWP distribution match the observations better, but the model produces too many thin clouds (low-LWP) and too few clear-sky (zero-LWP) cases (LCF≈ 45%). When the aerosol concentration is reduced further to *DM10 n_aer=0.1*, the LCF is clearly overestimated (∼ 69%). This sensitivity test suggests that reducing the assumed aerosol concentration in the deposition rate to observed levels can help to reduce the model's underestimation of cloud liquid water in winter.

The result that a reduction of the deposition rate helps to maintain liquid water in Arctic winter in the IFS is in line with results from Engström *et al.* (2014). They describe that reducing the deposition rate to 20% improves the representation of downwelling long-wave radiation (LW$_d$) in the ECMWF OpenIFS version (Cycle 36r4) compared with winter data from the SHEBA campaign. We find that a scaling of the deposition to 30% is sufficient to overcome the underestimation of LWP (LCF = ∼ 25%) during MOSAiC December (not shown).

IWP distributions

Reducing the deposition rate reduces the growth of ice, so we could expect a reduced vertically integrated ice water content (ice water path, IWP). However reducing the deposition rate can increase mixed-phase cloud lifetime, which can also increase the IWP. We find that the magnitude of changes in IWP to changes in the deposition rate is small compared with changes in LWP (Figure 5). The IWP increases when reducing the aerosol concentration from *DM10 n_aer=10* to *DM10 n_aer=1* (the median IWP changes from ~ 4 to $\sim 7\ \mathrm{g \cdot m^{-2}}$). When the aerosol concentration is reduced further to *DM10 n_aer=0.1*, the IWP starts to decrease (median IWP $\sim 5\ \mathrm{g \cdot m^{-2}}$). Part of the IWP is contributed by higher ice-only clouds, which do not respond to changes in the deposition rate. However, almost half of the ice mass in the simulations is located in lower levels (below 3 km), where the deposition rate is active. In total, the percentage of cloud ice below 3 km varies between 44% and 49%. The weak response can be explained by the competing effects of the deposition rate on ice, as it reduces the growth rate of ice from liquid but increases the lifetime of liquid cloud.

Cloud-base temperature and height

We reported a lack of cold clouds (below $\sim -20^{\circ}$C) in the model and an underestimate of near-surface (below 200 m) and higher clouds (above 1000 m) in winter, which we attributed (mostly) to temperature (see Section 3.1.4). Does the change in deposition rate improve those biases? In our winter SCM simulations with aerosol-aware INP parametrisation, we find that the occurrence of cold clouds increases when we reduce the assumed aerosol concentration (Figure 6, second column). In our optimised INP setting (*DM10 n_aer=2*), the occurrence of cold clouds (below $\sim -20^{\circ}$C) is even slightly overestimated. This overestimate indicates that the temperature dependence of the deposition rate in our optimised setting is not perfectly calibrated, now underestimating the deposition rate at cold temperatures. Clouds near the surface (below 200 m) and at heights above 600 m occur more frequently with decreasing aerosol concentration, which is where the colder clouds occur in the observations (see Figure 4 for DJF).

We conclude that targeting the temperature dependence of the deposition rate also targets the bias in cloud-base height occurrence in winter. Our optimised INP setting clearly improves the temperature dependence of the cloud bias, but, from our limited sample, seems to overcompensate slightly, probably underestimating the deposition rate at cold temperatures (below $\sim -20^{\circ}$C).

Surface downwelling long-wave radiation

The presence of liquid cloud water can increase the downwelling LW radiation emitted towards the surface, which is one main motivation for improving the representation of clouds in the Arctic. Arctic winter net LW radiation typically shows a bimodal distribution, with a lower peak for clear skies and a higher peak for cloudy conditions, which models struggle to reproduce (Pithan *et al.*, 2016; Stramler *et al.*, 2011). Here, we focus on downwelling LW radiation to investigate the atmospheric signal without the upwelling surface signal. Figure 6 shows the impact of the microphysics sensitivity experiments on ($\mathrm{LW_d}$), based on the month-long SCM simulations from December during MOSAiC. As shown in the previous section, the simulations differ in the occurrence of cloud liquid water, with the relative frequency increasing from $\sim 8\%$ to $\sim 45\%$ to $\sim 69\%$ when reducing the assumed aerosol concentration *n_aer* in the deposition rate from $10\ \mathrm{cm^{-3}}$ to $1\ \mathrm{cm^{-3}}$ to $0.1\ \mathrm{cm^{-3}}$. The increase in occurrence of cloud liquid water leads to an increase in ($\mathrm{LW_d}$) (Figure 6, first column). The control simulation and the *DM10 n_aer=10* simulation both underestimate the occurrence of cloud liquid water and the surface LW down radiation is underestimated by the model with mean errors of -19.9 and $-15.8\ \mathrm{W \cdot m^{-2}}$, respectively. Reducing the assumed aerosol concentrations increases the occurrence of larger LW down radiation. When the mean level of aerosol load is used (*DM10 n_aer=2*), the LW down radiation is represented best with a mean error of $-0.6\ \mathrm{W \cdot m^{-2}}$ (-0.3% relative error with a mean observed LW down radiation of $192.0\ \mathrm{W \cdot m^{-2}}$). If aerosol load is reduced further (*DM10 n_aer=0.1*), the LW down radiation is clearly overestimated with a mean error of $20.1\ \mathrm{W \cdot m^{-2}}$.

It is worth noting that the experiment *DM10 n_aer=2* with the lowest mean error in downward LW radiation also showed the best match to the frequency of liquid-containing clouds (LWP $>20\ \mathrm{g \cdot m^{-2}}$; see previous section). For further discussion, we will refer to the *DM10 n_aer=2* setup as the "optimised INP setting".

Now focusing on the distribution of downward LW radiation, the observations show a bimodal structure, where the lower peak ($\sim 160\ \mathrm{W \cdot m^{-2}}$) is attributed to clear-sky scenes and the higher peak ($\sim 230\ \mathrm{W \cdot m^{-2}}$) is attributed to cloudy scenes. In line with this attribution, the control experiment, which underestimates the occurrence of liquid-containing clouds, also underestimates the cloudy peak and overestimates the clear-sky peak in the LW down distribution, while the experiment *DM10 n_aer=0.1*, which overestimates the occurrence of liquid-containing clouds, also overestimates the cloudy peak and underestimates the clear-sky peak in the LW down distribution. The optimised INP setting experiment (*DM10 n_aer=2*) shows a bimodal structure, but the peaks are less well separated. This is consistent with the more frequent occurrence of intermediate values of LWP in the simulation compared with the observations (less frequent occurrence of zero LWP and high LWP values), as shown in Figure 5.

The LW down radiation emitted by a cloud depends not only on its phase and LWP but also on cloud-base temperature (Shupe & Intrieri, 2004), so we also compare the cloud-base temperature of observed and simulated liquid-containing clouds. Figure 6 shows the frequency of occurrence of liquid-containing clouds as

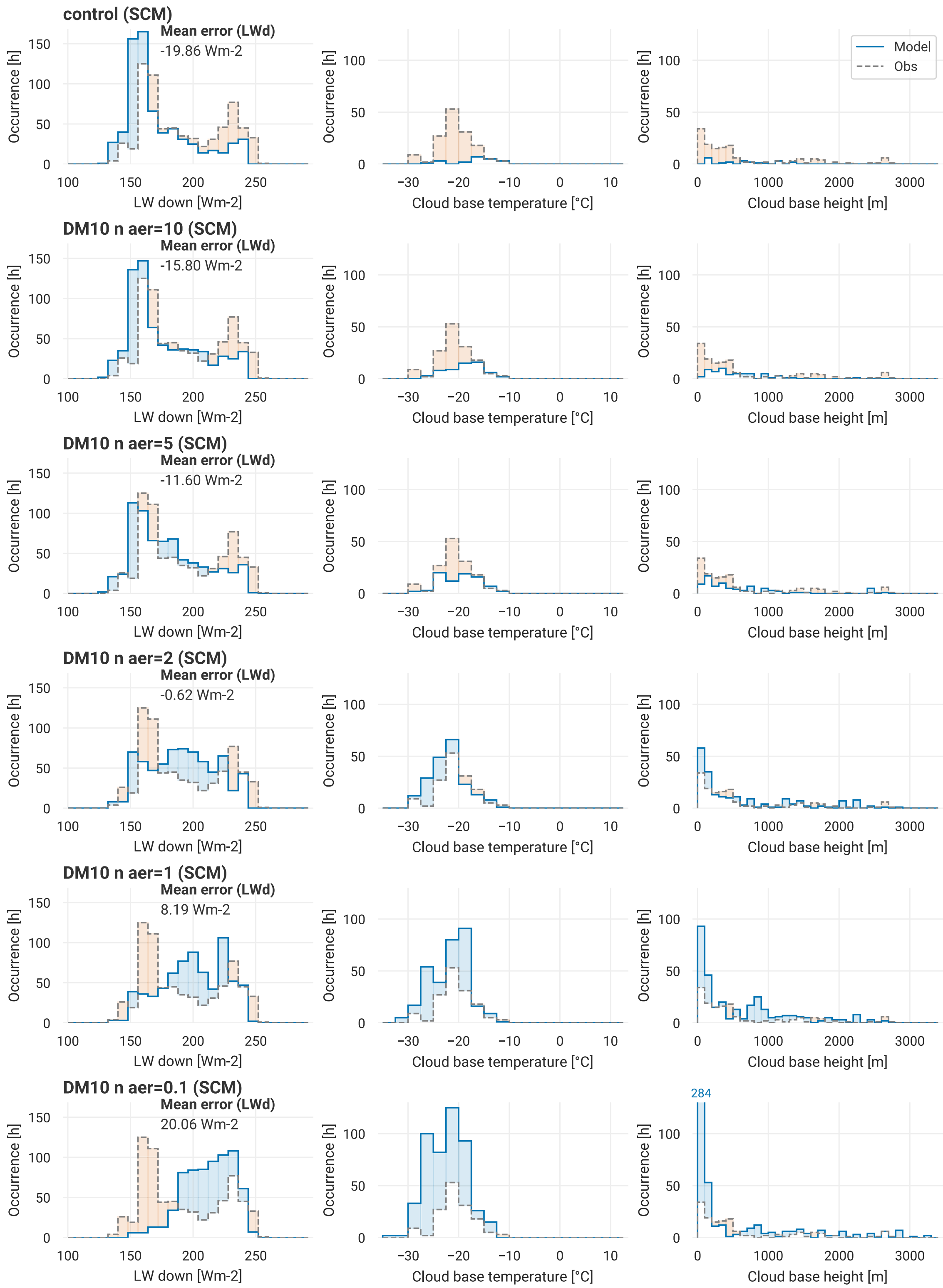

FIGURE 6 Occurrence frequency (hours) of surface radiation and cloud properties during MOSAiC in December for different SCM sensitivity experiments compared with the observations. Histograms of LW down radiation (first column). Histograms of the occurrence of liquid-containing clouds (LWP>20 $g \cdot m^{-2}$) with specific cloud-base temperature (second column) and cloud-base height (third column). Different rows show different SCM sensitivity experiments (experiment name given at top left). In all plots, model values are shown as solid lines, observations as dashed lines. The difference between model and observations is highlighted by a light colour shading: blue (red) for an overestimated (underestimated) occurrence in the model. Mean errors in LW down radiation (model-obs) are annotated in the first column plots. Note the different *y*-axis scales in the second and third columns. Values exceeding the axis limit are annotated above the bar. [Colour figure can be viewed at wileyonlinelibrary.com]

a function of cloud-base temperature and cloud-base height.

The optimised INP setting experiment represents the total frequency of liquid-containing clouds well, but has more colder cloud bases compared with the observations (Figure 6; fourth row, second column).

A colder cloud-base temperature leads to less LW down radiation, which can explain the shift of the cloudy peak in the LW down distribution to lower values (from ~ 230 W $\cdot$ m^{-2} in the observations to 215 W $\cdot$ m^{-2} in the model).

The maintenance of the supercooled liquid cloud water depends on the balance of sink and source processes. When we lower the deposition rate, we reduce a liquid cloud water sink and hence allow liquid water that is produced to be maintained more easily. The fact that the sensitivity experiments with reduced deposition rate maintain more liquid-containing clouds shows that source processes for cloud liquid water are active. We observe that experiments with strongly reduced deposition rate (*DM10 n_aer=2*, *DM10 n_aer=1*, and *DM10 n_aer=0.1*; still within the observed aerosol range) produce a lot of clouds close to the surface (cloud-base height < 100 m). The surface is mostly covered by sea ice, but in the case of open ocean, it can provide a moisture source for cloud formation. The sensitivity of the simulated clouds to sea-ice cover is investigated further in Section 3.3.2.

In conclusion, we showed that, by using reduced aerosol concentrations in the deposition rate, we can improve the representation of surface radiation in winter in the central Arctic. The improved surface radiation could lead to better predictions of surface temperature, particularly alongside surface scheme developments (Arduini *et al.*, 2022), highlighting the importance of representing liquid water in winter clouds accurately over the central Arctic.

3.2.2 | Case study: boundary-layer profiles

The presence of a liquid cloud layer can transform the boundary-layer temperature structure by cooling at the cloud top, inducing turbulent mixing below the cloud layer and warming the surface by long-wave radiation. Since the clouds may be maintained for several days, this can impact the temperature structure of the boundary layer and the evolution of sea-ice temperatures in the model.

Here, we present a case study of the mixed-phase cloud observed on December 17, 2019 at 1100 UTC. The cloud persisted for several days and was completely missed in the control forecast (see also Figure 1). We compare the representation of cloud liquid and ice water content, specific humidity, and temperature. As a reference we use a profile from a cloud remote-sensing retrieval, which scales the LWP between cloud boundaries and an interpolated radiosonde–meteorological tower product. We assess the control experiment and the sensitivity experiment with the optimised INP setting (*DM10 n_aer=2*) in the SCM setup and in the full 3D model (Figure 7).

The observed *cloud water content* profiles show a low liquid cloud layer (cloud top below 1000 m) and ice below the liquid layer. In the control setup, the SCM and 3D model simulations do not have any cloud liquid water in the lowest 2000 m and have very little cloud ice close to the surface. The 3D simulation with the optimised INP setting shows a clear liquid cloud layer with water content comparable with that of the observations. The corresponding SCM simulation also produces a liquid layer with comparable water content at slightly lower height. The height of the cloud layer is represented reasonably well in both simulations (100–200 m higher or lower than the observations, which is just the depth of one or two vertical model layers). The modelled ice water content in these simulations is of a similar order of magnitude to the retrieved cloud ice content, which is a significant improvement compared with the control simulations.

The relation of cloud properties and surface radiation is summarised in Table 1. LW down radiation is clearly underestimated in the control simulations by about −29% and −33% in the 3D simulation and SCM, respectively. In the simulations with optimised INP setting, where LWP is within the observed range (33 ± 20) g $\cdot$ m^{-2}, the relative error in LW down radiation is reduced to −9% and −8%, respectively. The strong relation between LWP and LW down radiation for LWPs (below 30 g $\cdot$ m^{2}) is in agreement with results from Shupe and Intrieri (2004).

The *temperature* profile in the control experiments shows a different boundary-layer structure from the observations. The observations show a temperature inversion at the cloud top, while the model experiments show a temperature inversion at a lower height above an approximately isothermal layer above the surface. The model near-surface temperature is about 5 K too cold and at the height of the observed cloud layer the model temperature is up to 10 K too warm (5 K in the SCM). In the experiments with the optimised INP setting, the temperature structure of the boundary layer shows a clear signature of the liquid cloud layers, with temperature inversions above the cloud layer in both SCM and 3D experiments, which is a clear improvement. In the 3D model experiment, in which the liquid layer is well represented, the temperature profile agrees very well with the observations. The large temperature errors that we described in the control experiment are reduced to almost zero. Only in the uppermost layer of the model cloud does a large temperature error remain, due to the inversion height being shifted by one model level. In the SCM, the temperature profile

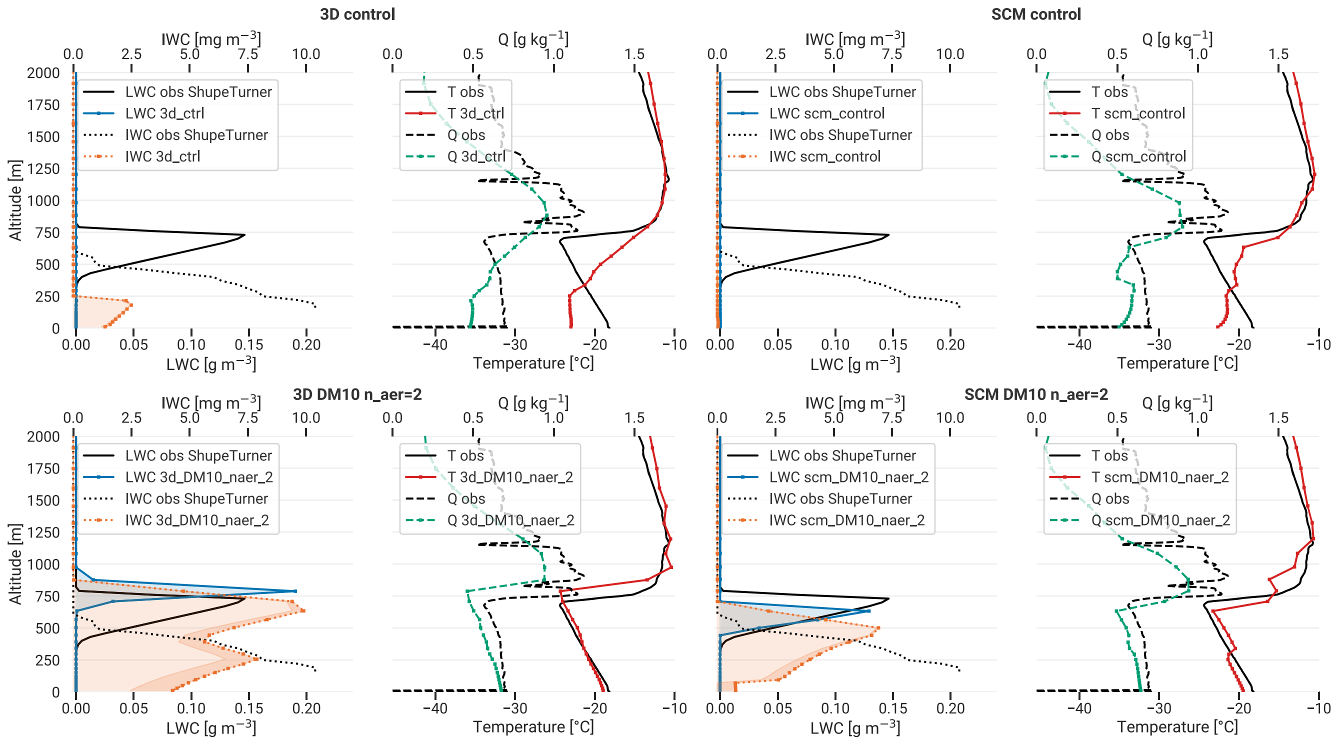

FIGURE 7 Impact of a reduced deposition rate on thermodynamic profiles in a case study from December 17, 2019, 1100 UTC. Boundary-layer profiles of liquid water content (LWC), ice water content (IWC), specific humidity (Q), and temperature (T) are shown for the control setup (top row) and the sensitivity test with a reduced deposition rate based on DM10 and an aerosol concentration of 2 cm^{-3} (bottom row). Profiles from 3D (left two columns) and SCM (right two columns) setups are shown. For reference, observed and retrieved profiles are included in black. Light orange shading indicates water content in the cloud ice category, dark orange shading indicates water content in the snow category. [Colour figure can be viewed at wileyonlinelibrary.com]

TABLE 1 Summary of cloud properties and surface radiation for the case study from December 17, 2019, 1100 UTC (hourly average). For the simulation *3D DM10 n_aer 2*, cloud-base temperature and height are given in brackets, because LWP is below 20 g · m^2.

	LWP	Cloud-base temperature	Cloud-base height	LW down
Obs	33.2 g · m^2	−22.4°C	456 m	229.2 W · m^{-2}
3D ctrl	0.0 g · m^2	–	–	160.7 W · m^{-2}
SCM ctrl	0.0 g · m^2	–	–	154.2 W · m^{-2}
3D DM10 n_aer 2	19.7 g · m^2	(−24.0°C)	(708m)	209.3 W · m^{-2}
SCM DM10 n_aer 2	16.8 g · m^2	(−21.9°C)	(500.9m)	211.4 W · m^{-2}

shows inversions at the top of both modelled cloud layers. Although the structure of the temperature profile does not align with observations, the temperature error at the surface is significantly reduced, with a residual cold bias of about 2 K.

The observed *moisture* profile shows a clear humidity inversion above the cloud layer. These moist layers often occur in the Arctic and are a potential moisture source for the liquid cloud layer (e.g., Morrison *et al.*, 2012). The humidity profile is approximately captured in the model control runs but is much smoother and misses the sharp humidity inversion seen in the observations, and the fine-scale structure of moist layers above the inversion is not resolved. In the *n_aer=2* experiments, where the cloud profile is much closer to the observations, the sharpness of the humidity inversion is much better represented.

To conclude, we clearly see how the reduced INP concentration in the deposition rate improves the representation of the boundary-layer structure and the near-surface temperature in this case study substantially. However, predicting the time and height of a cloud layer depends on other aspects of the atmospheric state, so we cannot expect to see this improvement for each scene. Nevertheless, the strong impact on boundary-layer structure and surface

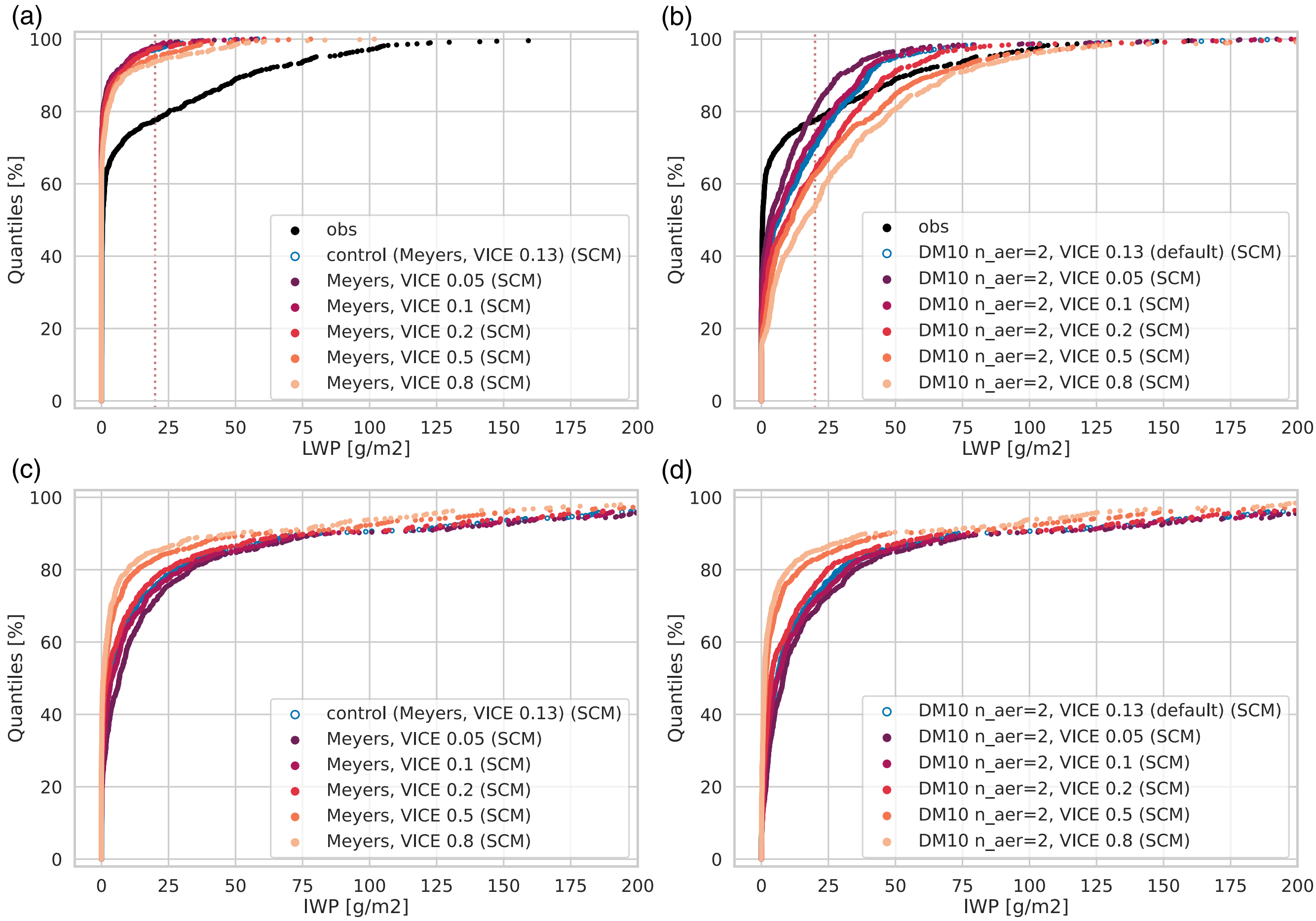

FIGURE 8 Sensitivity of modelled LWP and IWP to ice-particle fall speed during MOSAiC December. (a) LWP distribution and (c) IWP distribution using the control INP parametrisation (Meyers) and varying ice fall speed. (b) LWP distribution and (d) IWP distribution using the "optimised INP setting" (*DM10 n_aer=2* INP parametrisation) and varying ice fall speed; each point represents an hourly averaged value. [Colour figure can be viewed at wileyonlinelibrary.com]

temperature is a motivation for adapting the microphysics for the central Arctic to improve the representation of cloud liquid water.

3.3 | Sensitivitiy of LWP to other processes in winter Arctic clouds

We showed that the model cloud liquid water in winter is very sensitive to the deposition rate. We now examine how other uncertain processes, such as the ice-particle fall speed and the sea-ice fraction, affect the representation of liquid and ice water in winter Arctic clouds.

3.3.1 | Sensitivity to ice-particle fall speed

We expect that increasing the ice-particle fall speed has the potential to increase LWP, as it can help separate liquid and ice in a liquid-topped mixed-phase cloud. We test the impact of varying the fall speed between 0.05 and 0.8 $\mathrm{m \cdot s^{-1}}$ on LWP and IWP (the default ice fall speed is 0.13 $\mathrm{m \cdot s^{-1}}$; see Figure 8). We find that varying the ice fall speed (in the given range) has a limited effect on the LWP distribution, if we use the default Meyers INP parametrisation (LCF between $\sim 2\%$ and $\sim 6\%$). The sensitivity is increased when we use the optimised INP setting (*DM10 n_aer=2*; LCF between $\sim 20\%$ and $\sim 46\%$). We find a reduction in IWP when the ice fall speed is increased, independent of the INP parametrisation used.

The dependence of the fall speed of ice particles on its size and shape is an active area of research; our tested values fall broadly in the range found by measurements (e.g., in Kiruna by Vázquez-Martín *et al.*, 2021). The plan is to use a diameter (mass-weighted) fall speed in future versions of the IFS. The sensitivity tests show that, while the details of the LWP distribution can be expected to change with fall speed, this does not alter the overall conclusions regarding INP assumptions about the deposition rate in winter Arctic mixed-phase clouds.

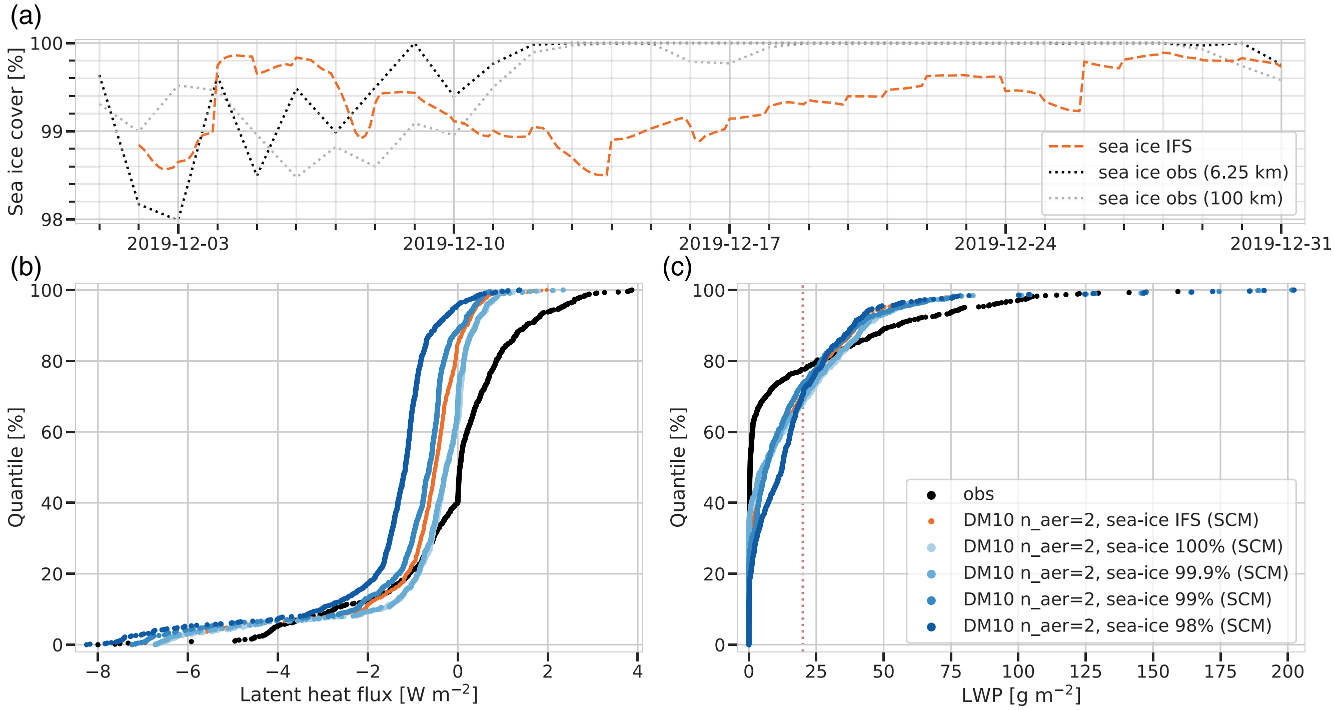

FIGURE 9 Sensitivity of modelled LWP to sea-ice cover during MOSAiC December. (a) Time series of observed sea-ice cover from satellite retrievals with 6.25-km and 100-km spatial resolution (dotted lines). (b) Surface latent heat flux (positive downward) and (c) liquid water path distribution in the SCM sensitivity experiments and from observations, respectively; each point represents an hourly averaged value. Negative values of latent heat flux indicate evaporation from the surface. [Colour figure can be viewed at wileyonlinelibrary.com]

3.3.2 | Sensitivity to sea-ice cover

Moisture fluxes from open leads in the sea-ice are a source of moisture that can impact the cloud cover (Section 2.1.1). Figure 9a shows the sea-ice cover in the model and observed around *Polarstern* by a satellite retrieval (Section 2.2). The model sea-ice cover is underestimated in December: the observed sea-ice cover increases from 98% at the beginning of December to close to 100% on December 12, while the model keeps up to 1.5% open ocean and only reaches concentrations close to 99.9% on December 26, two weeks later than observations. By design, the modelled sea-ice cover never reaches 100% sea-ice cover, because an upper threshold of 99.9% is applied as a design choice in the sea-ice model. Because we expect more evaporation from open ocean compared with full sea-ice cover, we aim to determine whether small deviations from 100% sea-ice cover influence the representation of liquid clouds.

To explore this sensitivity, we prescribe fixed sea-ice cover levels in the SCM of 98%, 99%, 99.9% and 100% for the month of December. We run the experiments with optimised INP setting *DM10 naer 2* (see discussion in Section 3.2), where we see a stronger sensitivity than in the control setting. In the control, the LWP shows no sensitivity to increased sea-ice cover, as LWP is mostly zero and an increase in sea-ice cover can only reduce the already rare occurrences of non-zero LWP. The reduction to a fixed sea-ice cover of 98% is not enough to increase the LWP (not shown).

The sea-ice sensitivity experiments show that the change of sea-ice cover is accompanied by a change in surface evaporation, visible through the surface latent heat flux (Figure 9b). Experiments with higher sea-ice cover show a reduction in the occurrence of negative (upward) latent heat fluxes, that is, less surface water evaporates. Compared with observations, experiments overestimate the occurrence of strong negative latent heat flux ($< -5\ \mathrm{W \cdot m^{-2}}$) and underestimate the occurrence of moderately positive latent heat flux ($> 1\ \mathrm{W \cdot m^{-2}}$). It is worth noting that the surface flux observations are taken above sea-ice and their narrow footprint could not capture fully the influence of small leads that may be present within the area represented by a larger model grid box.

The sensitivity tests show a small influence of sea-ice cover on LWP, as illustrated in Figure 9c. We see an increase in the occurrence of zero LWP when high sea-ice cover is prescribed (100% or 99.9%), which is a slight improvement compared with the experiment where IFS sea-ice cover is used. With 1% or 2% of open ocean, the occurrence of thin clouds ($<20\ \mathrm{g \cdot m^{-2}}$) increases.

To summarise, the simulated cloud liquid water shows a small sensitivity to small fractions of the open ocean when the deposition rate is reduced, which we expect in clean atmospheric conditions. The SCM results indicate that, if we want to constrain the deposition rate based on the observed LWP of MOSAiC, the exact representation of sea-ice cover in the model becomes more important. Further 3D model tests on the deposition rate could be run with observed sea ice initially, rather than coupled, since errors grow quickly in the medium range (Day *et al.*, 2022).

4 | DISCUSSION

Our results show a strong sensitivity of LWP to different INP concentrations. Is this strong dependence on INP concentration realistic or does the structure of the parametrisation amplify the dependence? In the following, we discuss some design choices in the parametrisation regarding their potential to change the sensitivity to INP.

- *Fixed ice number concentration.* We use the INP concentration from the DeMott *et al.* (2010) parametrisation directly for a fixed diagnostic ice number concentration. This can represent the initiation of ice particles and their initial growth; however, not dynamically when ice nucleating particles are used up or recycled (e.g., Solomon *et al.*, 2009). Furthermore, secondary ice processes can increase the number concentration above INP levels by several orders of magnitude, which could reduce the impact of INP concentrations in determining the ice concentration and consequently the deposition rate (e.g., Pasquier *et al.*, 2022). Ice particles nucleated in cold layers can be transported to warmer environments and increase the number concentration there.
- *Ice particle characteristics.* Assumptions about the shape and size distribution of ice particles can impact the deposition rate and its sensitivity to the assumed ice number concentrations and mass (e.g., Ong *et al.*, 2024). Here, we assume spherical, monodisperse particles, following Rotstayn *et al.* (2000). The assumed mass–diameter relation influences how a change in number concentration affects the particle diameter for a given ice mass. Since the diameter relates to capacitance, it ultimately affects the deposition rate. By this the mass–diameter relation can affect the dependence of the deposition rate on ice particle number concentration.
- *Spatial heterogeneity.* Studies suggest that the WBF process is less active in mixed-phase clouds because of supersaturation with respect to ice and liquid in updraughts and subsaturation with respect to ice and liquid in downdraughts (Korolev, 2007; Korolev & Field, 2008). Consequently, depositional ice growth would only take place in the updraught parts of the grid box. The subgrid dynamics that would be needed to account directly for different saturation environments is not resolved, but is taken into account by reducing the deposition rate by a fixed scaling factor to represent the fraction of the grid box where the supercooled liquid and ice overlap. The spatial heterogeneity of cloud ice and liquid is highly uncertain (Korolev & Milbrandt, 2022). There could be a temperature dependence of the heterogeneity: for example, Sokol and Storelvmo (2024) use Cloud–Aerosol Lidar and Infrared Pathfinder Satellite Observations (CALIPSO) lidar data and show that the horizontal spatial heterogeneity of the cloud phase is greatest between −4°C and −15°C, with less heterogeneity (and therefore more overlap of liquid and ice) as temperatures decrease below this. This would increase the deposition rate with decreasing temperatures below −15 °C, and decrease the supercooled liquid water even further where we already have too little at cold temperatures. If this temperature dependence of the spatial heterogeneity was parametrised, the assumed INP temperature dependence should be weakened to keep the temperature dependence of the deposition rate, but there is still much uncertainty in this study and it is a topic for further work.
- *Competing microphysical processes.* Other processes could be responsible for converting liquid to ice, like riming of the falling ice particles or immersion freezing of larger droplets, but these processes are not accounted for directly in the current IFS implementation. Their dependence on INP concentration may be different from the INP dependence of the deposition process and thereby could possibly reduce the actual dependence on INP.

Our sensitivity tests suggest that incorporating aerosol concentration in the deposition rate can be a determining factor in predicting LWP in remote regions at cold temperatures. Incorporating a dependence on climatological aerosol concentrations could reduce the deposition rate in the Arctic without affecting more polluted areas. Our sensitivity tests also suggest a strong sensitivity of LWP to different aerosol concentrations within the observed range. The strong sensitivity of LWP within the range of observed values may suggest that the day-to-day variability of aerosol needs to be represented to simulate appropriate deposition process rates. Therefore, model experiments using aerosol concentrations from global aerosol fields would be the next step, but are beyond the scope of this study. Both global aerosol climatologies and aerosol forecasts rely on atmospheric composition models like the Copernicus Atmosphere Monitoring Service (CAMS:

TABLE 2 Observational data used for evaluation.

Parameter	Link
Liquid water path	https://doi.org/10.1594/PANGAEA.941389
Cloud-base height	https://doi.org/10.1594/PANGAEA.935263
	https://doi.org/10.1594/PANGAEA.935264
	https://doi.org/10.1594/PANGAEA.935265
	https://doi.org/10.1594/PANGAEA.935266
	https://doi.org/10.1594/PANGAEA.935267
Cloud liquid and ice water content	https://www.osti.gov/biblio/1871015
	(doi: 10.5439/1871015)
Interpolated radiosonde	https://doi.org/10.5439/1095316
Extended radiosonde profiles	https://doi.org/10.1594/PANGAEA.961881
Long-wave broad-band downwelling irradiance	https://doi.org/10.5439/1333228
Coarse aerosol concentration	https://doi.org/10.1594/PANGAEA.961068
Surface latent heat flux	https://doi.org/10.18739/A2PV6B83F
Sea-ice fraction	https://seaice.uni-bremen.de

Morcrette *et al.*, 2009. The reliability of aerosol number concentrations, especially in the Arctic, is unclear. However, the work from Dietel *et al.* (2024) suggests that aerosol concentrations from the CAMS reanalysis (Inness *et al.*, 2019) have predictive power for the cloud phase as observed by satellites. Further research is needed to assess the benefit of coupling the deposition rate to aerosol data in a global numerical weather prediction context.

5 SUMMARY AND CONCLUSION

This study investigates the representation of liquid-containing clouds in the ECMWF IFS model during the full year of Arctic ice-drift campaign MOSAiC. The model underestimates the occurrence of liquid-containing clouds with cold cloud-base temperatures (below −15°C) and overestimates the occurrence of liquid-containing clouds with warmer cloud-base temperatures (above −5°C). The frequency of occurrence of liquid-containing clouds in the model in cold temperatures increases when the ice depositional growth rate is reduced. A reduced deposition rate can be expected with lower concentrations of ice-nucleating particles (INPs) in the clean Arctic atmosphere, as we show by using a simple aerosol-aware INP parametrisation by DeMott *et al.* (2010) with aerosol levels in the range of the observed values. Such an adjustment can improve the representation of surface radiation significantly, reducing the LW down bias from −19.9 to $-0.6\,\mathrm{W \cdot m^{-2}}$, as shown in model simulations in a single-column configuration over one winter month (December). It also has the potential to improve the boundary-layer temperature and moisture profiles and near-surface air temperature, if the time, height, and phase of a cloud are predicted correctly, which we highlight in a specific case study in both the SCM and the full global model.

We tested the sensitivity of the occurrence of liquid-containing clouds to other processes. While changes to ice-particle fall speed can affect the occurrence and water content of liquid-containing clouds, an increase in ice-particle fall speed alone cannot resolve the underestimation in Arctic winter. When the deposition rate is reduced, the sensitivity to ice-particle fall speed increases.

Lastly, we find that sea-ice cover influences both the occurrence and water content of liquid-containing clouds in the IFS in winter SCM experiments. However, the impact of small deviations from the observed full sea-ice cover is relatively minor compared with the model's sensitivity to the deposition rate.

We conclude that reducing the deposition rate can clearly help to improve the underestimation of Arctic mixed-phase clouds in winter in the model. The deposition-rate reduction can be achieved through the use of a comparatively simple aerosol-aware deposition rate, using the INP parametrisation by DeMott *et al.* (2010) with observed aerosol levels. The feasibility of using aerosol concentrations from global data sets in the Arctic remains to be tested. Furthermore, more sophisticated INP parametrisations that include speciation of aerosol could increase the realism of INP predictions, but have not been tested in the IFS. With the help of the comprehensive

MOSAiC dataset, any new setup of INP parametrisation and aerosol data should be tested by evaluating the occurrence of liquid-containing clouds in different temperature regimes.

The availability of observations in the Arctic ocean region in winter is sparse, as most of the ground-based observations from ship campaigns in the central Arctic are in summer. While this study has focused primarily on improving the representation of supercooled cloud liquid water at cold temperatures, addressing the model's overestimation of liquid-containing clouds with warm cloud-base temperatures near −5°C remains an important avenue for future work. Since model developments can affect clouds across seasons, evaluating Arctic clouds over the full annual cycle is essential to ensure robust improvements in their representation.

Since the IFS is a global model, it will be necessary to assess the impact of Arctic-suiting parametrisation developments in a broader region. Liquid cloud detection from satellite observations such as those from *CloudSat*/CALIPSO (Delanoë & Hogan, 2010) or the recently launched *EarthCARE* satellite (Wehr *et al.*, 2023), which cover the globe up to ∼ 82°N/S, could help to increase coverage of the temperature-dependent liquid cloud error analysis.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The single-column model forcing data for the month of December for MOSAiC, the SCM experiment output, and the extracted MOSAiC track from the 3D IFS forecasts (day 2) are available at https://doi.org/10.5281/zenodo.17978089. The single-column model forcing data will be made available in DEPHY format on the github repository (https://github.com/GdR-DEPHY/DEPHY-SCM) for use in other single-column models.

Observational data used for evaluation are publicly available and are summarised in Table 2.

ENDNOTE

[1] https://climate.copernicus.eu/esotc/2020/arctic-temperatures.

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